CONFIDENTIAL **MOCK EXAM 2026**

GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A (80 MARKS)

QUESTION 1

a) Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
KLCPJLYLABYULKHUKDPWLKVUL

(2 marks)

b) Explain substitution ciphers by describing how plain-text is transformed, what serves as the key, how the approach relates to Shannon's principle of confusion, and how this idea carries over to modern ciphers.

(6 marks)

QUESTION 2

a) State TWO (2) typical locations where IDS sensors are placed in a network-based IDS architecture.

(2 marks)

b) Describe six (6) preventive network security mechanisms that an organization can deploy to protect its network, explaining what each mechanism does.

(6 marks)

QUESTION 3

a) State TWO (2) facts about the chain of trust in Trusted Computing, focusing on how it begins and how measurements are recorded.

(2 marks)

b) Explain the roles of the two primary system components of the Next Generation Secure Computing Base, describing what each component is and how it supports trusted execution on the platform.

(6 marks)

QUESTION 4

a) List the TWO (2) stages that make up the biometric authentication process.

(2 marks)

b) State the definition of a biometric as given in the lecture, and state the TWO (2) sectors into which today's biometric applications are categorised. Then name the THREE (3) biometric technologies listed under the behavioural sector.

(6 marks)

QUESTION 5

a) Explain two characteristics of Trojan horse malware, including how it disguises itself and what harm it can cause once active on a host.

(2 marks)

b) Discuss six ways in which malware attacks and propagates, covering how malicious code is attached to executables, how early viruses and worms spread, how rootkits and spyware operate on infected hosts, and how self-replication supports propagation.

(6 marks)

QUESTION 6

a) Explain the difference between morals, ethics and laws as three forms of guidance for conduct in computer security work.

(2 marks)

b) Explain six requirements that a professional code of ethics and conduct, such as the ACM code, places on a computing professional's behaviour in security-related work.

(6 marks)

QUESTION 7

a) Explain the Chinese Wall model as a formal access-control policy, stating the specific concern it addresses and how its access decisions depend on a subject's access history.

(2 marks)

b) Compare the access control matrix, access control lists and capability lists as structures for implementing access policies, and state the requirements that any such access control structure must satisfy.

(6 marks)

QUESTION 8

a) Explain what a buffer overflow is and how it can be exploited to hijack a program's execution flow.

(2 marks)

b) State six Windows security mechanisms or components and, for each, briefly indicate the security function it performs.

(6 marks)

QUESTION 9

a) State two classes of threats to a computer system and, for each, identify the security objective whose violation represents the harm caused.

(2 marks)

b) Describe the Samy worm as a case study of a cross-site scripting attack, covering how the malicious script was injected, how it spread to other users, and the outcome of the incident.

(6 marks)

QUESTION 10

a) List two ways in which an intrusion detection system can support the detection and defence against attacks on a computer system.

(2 marks)

b) Refer to your answer in (a). Explain how the distinction between a threat and an attack, together with the role of vulnerabilities, shapes the countermeasures a security team should deploy against malware and other threat classes.

(6 marks)

PART B (20 MARKS)

QUESTION 1

A software house is drafting a code of ethics for its security team. A junior developer asks why the team cannot simply rely on the ACM code, and a manager wants to know what such a documented code can and cannot achieve for the group.

a) Explain what a documented code of ethics provides to a professional group, and discuss the limitations such a code has when guiding the ethical conduct of its members.

(5 marks)

b) Referring to the scenario, explain five ethical concerns the security team's code of ethics should address when handling personal information, and why the organization itself, not only individual developers, must take responsibility for protecting that information.

(5 marks)

QUESTION 2

A system administrator is reviewing how the Linux servers in a campus laboratory protect user accounts. She wants to strengthen the controls applied to password files and login behaviour, and is also concerned about how passwords are stored in encrypted form.

a) State FIVE (5) password security controls that can be applied to protect user accounts, and state how a password is stored in encrypted form.

(5 marks)

b) Referring to the scenario, the administrator considers deploying Kerberos and single sign-on so that laboratory users authenticate once and are not burdened by repeated password entry. Discuss the motivation for single sign-on and the design criteria that Kerberos must satisfy, and explain how a user's password is used at login.

(5 marks)

END OF QUESTION PAPER